

ADEENA RAMZAN

AI/ML Engineer | Generative AI & RAG Systems | LLM Fine-tuning

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PROFESSIONAL SUMMARY

AI/ML Engineer with production deployments spanning Generative AI, cybersecurity, medical imaging, healthcare analytics, and environmental AI. Skilled in building and deploying RAG pipelines with LLM integrations (ChatGPT, Claude, Gemini, Llama, Groq), fine-tuning Transformer models (CodeBERT, BERT, BiLSTM), and shipping production REST APIs via FastAPI and Docker. Experienced with AI-assisted agentic development tools (V0, Lovable, Antigravity) and end-to-end MLOps from data pipelines and training to live deployment on HuggingFace Spaces, Vercel, and Streamlit.

WORK EXPERIENCE

Machine Learning Engineer

08/2026 – Present

Syslab, Karachi

- Working on machine learning, deep learning, and LLM-based systems in a production engineering environment.

AI Research Intern | Water Chemistry

02/2026 – 06/2026

Alphatron Technologies (Pvt.) Ltd., in collaboration with NED University, Karachi

- Engineered AI-powered data pipelines for industrial water monitoring: designed system architecture, built WQI computation workflows, and trained classification, regression, and time-series forecasting models on real-world field data.
- Integrated LangChain LLM (Gemini, Llama) with a Vector Database for a RAG-based query interface on the trained classifier; researched Chain-of-Thought and Tree-of-Thoughts prompting for production Generative AI workflow design.

AI Trainee Intern

10/2025 – 12/2025

Wenawa, Hybrid

- Fine-tuned ResNet-50 for 5-class skin disease detection achieving **78.3% validation accuracy** via 3-stage training, custom augmentation, and hyperparameter optimization; deployed Grad-CAM inference app on HuggingFace Spaces via Gradio.
- Built Sindhi-to-Urdu/English translation tool using Google Translate API and Express.js backend, extending platform accessibility across 3 language communities.

IT Instructor (Part-time)

03/2026 – 05/2026

Benazir Bhutto Shaheed Youth Development Program, Karachi

- Delivered AI, ML, and Web Dev curriculum to 20+ students per cohort for a government-backed digital skills initiative; designed 8+ hands-on lesson plans.

SKILLS

- Languages & ML:** Python, JavaScript, TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM, Flask, CNNs, ResNet, Transformers, CodeBERT, BERT, BiLSTM
- Generative AI:** ChatGPT, Claude, Gemini, Llama, Groq, LangChain, RAG, FAISS, ChromaDB, Vector DBs, LLM Fine-tuning, Prompt Engineering
- Backend & Frontend:** FastAPI, WebSockets, SQLAlchemy, SQLite, PostgreSQL, Docker, React, Next.js, Node.js, Streamlit, Gradio
- Deployment & Tools:** HuggingFace Spaces, Vercel, V0, Lovable, Antigravity, Pandas, NumPy, EDA, Git

PROJECTS

SecureScope AI: AI-Powered Python Vulnerability Scanner

[GitHub](#) | [Demo](#)

Python, TensorFlow/Keras, PyTorch, XGBoost, LightGBM, CodeBERT, FAISS, FastAPI, Docker, HuggingFace, Groq, Llama

- Engineered a 22-feature vulnerability detection pipeline using AST structural analysis and regex pattern matching; trained an ANN, XGBoost, and LightGBM ensemble on 14,248 real Python functions, achieving F1 0.723 on a held-out benchmark of 3,563 real-world GitHub functions.
- Trained a Bidirectional LSTM on raw token sequences, improving F1 to **0.9505**; fine-tuned `microsoft/codebert-base`, achieving **F1 0.954** and **ROC-AUC 0.989** on the same benchmark without relying on hand-crafted features.
- Integrated a hybrid CodeBERT + FAISS RAG pipeline over OWASP documentation with Groq Llama 3.1 8B Instant for automated fix suggestions; deployed as a multi-mode FastAPI REST API, containerized with Docker on Hugging Face Spaces with a Vercel frontend.

Enterprise Employee & Endpoint Activity Monitoring System

[GitHub](#)

Python, FastAPI, WebSockets, SQLAlchemy, SQLite, Asyncio, Win32 API (ctypes), MSS, PyInstaller, Tkinter

- Built a full-stack real-time IT endpoint monitoring and remote-support platform enabling admins to audit activity, enforce policy, and provide interactive remote assistance across a LAN via a FastAPI/WebSocket backend and a glassmorphic JS dashboard.
- Implemented low-latency (20+ FPS) remote screen streaming with full mouse/keyboard control and one-click remote screen lock using low-level Win32 input hooks for IT maintenance workflows.
- Engineered activity and DLP logging (app tracking, USB events, file transfers, browser history) with idle-time detection and policy alerting; compiled a zero-dependency standalone Windows client agent (PyInstaller) with automatic LAN backend discovery.

Care4Diabetics Extension: Diabetic Complication Prediction (FYP)

[GitHub](#)

Python, LightGBM, Random Forest, Scikit-learn, Flask, Pickle

- Developed two binary classification models extending Care4Diabetics, in collaboration with DOW University Hospital: LightGBM for retinopathy (**ROC-AUC 0.9315**) and Random Forest for neuropathy (**ROC-AUC 0.9370**) on a real 1,500-patient Pakistani clinical dataset, validated via 50-fold cross-validation with zero data leakage.

Generative AI Systems — RAG Chatbot & Multi-Agent Research

[RAG Chatbot](#) | [Demo](#) | [Multi-Agent](#) | [Demo](#)

LangChain, ChromaDB, Groq LPU, FastAPI, SSE, React 19 — Built a document Q&A system retrieving answers exclusively from uploaded documents with source citations; orchestrated specialized AI agents with a modular multi-LLM resolver (Groq, Gemini, OpenAI, Ollama) for autonomous web research with real-time SSE streaming.

Additional ML Projects — Computer Vision, NLP & Business ML

[DermNet](#) | [Emotion](#) | [Churn](#) | [Heart](#)

- 3-stage ResNet-50 skin disease classifier (**78.3% acc.**, Grad-CAM, DermNet); text emotion classifier (**89.7% acc.**, TF-IDF); customer churn model (**F1 0.9943**, SMOTE); heart disease classifier (**83% acc.**) — all deployed as live Streamlit/Gradio apps.

HACKATHONS

SkillBridge: AI-Powered Multilingual Career Mentor

[Demo](#)

Kanz AI Skills Hackathon 2026 | React.js, Node.js, Gemini 1.5 Flash, LangChain, Stripe — Built a bilingual (Arabic/English) AI career mentor for Arab youth solo within the hackathon timeframe: personality-based career quiz, AI chat mentor, and mock interview simulator with real-time feedback.

EDUCATION

Bachelor of Computer Science (BSc)

02/2023 – Present

University of Karachi (UBIT)

Relevant Coursework: Statistics, Linear Algebra, Data Warehousing and Mining, Network Security, DSA, Simulation and Modelling, Software Engineering

CERTIFICATIONS

- AI Upskilling: Technical Foundations (*Qualcomm Academy*)
- Introduction to NLP Concepts (*Microsoft Learning*)
- Machine Learning Introduction for Everyone (*IBM*)
- Python for Data Analysis: Pandas and NumPy (*Coursera*)